Papuan V2 FER Dataset — Final Delivery Package

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1. What this dataset is

This package contains 231 face crops extracted from a single source video (**pesta_babi.mp4**), intended as a working facial-expression-recognition (FER) dataset for thesis research. Each sample is a cropped face image, available in two variants (a plain crop and a CLAHE-contrast-enhanced version), standardized to a fixed 224x224 pixel size. These 231 samples are the subset of 645 originally extracted crops that passed a basic quality-control filter (described in section 4).

2. Folder guide

Folder	Contents
02_Images/Original/	231 standardized (224x224) plain face crops, one JPEG per sample.
02_Images/CLAHE/	231 standardized (224x224) contrast-enhanced face crops, same geometry as Original.
03_Landmarks/Initial/	231 JSON files: 42-point facial landmark predictions (AI-generated or reused), in standardized 224
03_Landmarks/Reviewed/	26 JSON files: human-corrected 42-point landmarks for the subset that has been reviewed.
04_Manifests/	CSV spreadsheets describing every sample (see section 3).
05_Quality_Report/	QC methodology, identity-pilot report, example contact sheets, and this package's LIMITATIONS.m
00_README.pdf	This document.
01_Dataset_Overview.xlsx	Summary statistics and a plain-English data dictionary for every manifest column.

3. sample_id vs. person_id

sample_id (e.g. *frame_000010_face02*) identifies one detected face crop: which sampled video frame it came from and which detection within that frame. It is a stable, reliable key for joining images, landmarks, and labels.

person_id is NOT a reliable identity yet. No cross-frame identity tracking has been run on this dataset at scale. Every sample defaults to person_id = **"unknown"**, meaning identity was never assessed. A small 31-crop pilot test (covering only 5 consecutive sampled frames) tried matching faces across frames using AI face-embedding similarity; 11 of the 231 delivered samples were involved in a weak, unconfirmed match from that pilot and are marked person_id = **"needs_review"** instead. This is still not a confirmed identity — it only means a human should look at it. Two specific comparisons from that pilot were too ambiguous to call either way and are left explicitly unresolved. Do not use person_id for any identity-grouped analysis (e.g. train/test splits by person) without first running and validating a proper tracking/re-identification step.

4. 42-point landmark schema

Landmarks follow the standard 68-point facial landmark convention (as produced by the InsightFace landmark_3d_68 model), but only a 42-point subset relevant to expression is kept here, dropping the jaw outline and nose. Each point in the JSON files keeps its original 0-67 *source_index* so it can always be traced back to the full 68-point scheme.

Region	Source indices (0-67)	Count

Right eyebrow	17-21	5
Left eyebrow	22-26	5
Right eye	36-41	6
Left eye	42-47	6
Mouth (outer + inner)	48-67	20
Total		42

Coordinates in this delivery are in *standardized 224x224 pixel space* (i.e. already transformed to match the Original/CLAHE images in 02_Images/), not the original crop's native pixel space.

5. Quality control (QC)

Of 645 originally extracted face crops, 231 passed a basic quality filter: blur-variance (a sharpness measure) greater than 75, and a minimum crop dimension (width and height) of at least 60 pixels, plus requiring a successfully-generated CLAHE counterpart. These thresholds came from a small human-reviewed visual pilot, not a large validated classifier — see 05_Quality_Report/QC_THRESHOLD_RECOMMENDATION.md for the full methodology. The 414 excluded crops were too blurry or too small by this heuristic; they are not included in this package at all.

6. Completion status (231 delivered samples)

Aspect	Complete	Pending
Image standardization (both variants)	231 / 231	0
Initial 42-point landmarks	231 / 231	0
Human-reviewed landmarks	26 / 231	205
Expression label (single annotator)	52 / 231	179
person_id (validated identity tracking)	0 / 231	231 (by design, see section 3)

7. Explicit limitations

- Face bounding boxes were never saved by the original detection pipeline — only the final cropped image. See 05_Quality_Report/LIMITATIONS.md item 1.
- person_id is unvalidated at full-dataset scale; only a 31-crop pilot (5 frames) was tested. See section 3 and LIMITATIONS.md item 2.
- That same pilot found 6 crops in its 31-crop subset that are not real faces (upstream detector false positives); only this subset has been checked, not the full 231 or 645. See LIMITATIONS.md item 3.
- 205 of 231 samples have only AI-predicted (not human-reviewed) landmarks. See LIMITATIONS.md item 4.
- 179 of 231 samples have no expression label yet, and all existing labels are single-annotator only (no cross-annotator verification exists anywhere in this project). See LIMITATIONS.md item 5.
- The QC threshold used to select these 231 samples is pilot-derived, not a large-scale validated classifier. See LIMITATIONS.md item 6.

8. Full detail

See 05_Quality_Report/LIMITATIONS.md for the complete, citation-linked version of every item above, and 04_Manifests/annotation_checklist.csv for a per-sample to-do list of exactly what remains (missing label, missing landmark review, identity needs review).